

FNU MANSOOR
Mobile: (251) 391-1108
Email: mansoor7696@gmail.com

Summary:

- RF Engineer having experience in designing and testing of LTE and Ultra-wideband antennas
- Experience in working from 1 MHz to 110 GHz
- Worked on RF Antenna designing simulation tools such as CST Microwave Studio, HFSS, ADS
- Ability to test different parameters S-parameters, RF power, Noise floor, Passive Intermodulation, etc.
- Experience in Passive Intermodulation (PIM) on base station antennas
- Worked on amplifiers, filters, detector, equalizer, switches, coupler etc.
- Good knowledge of lab equipment such as Network analyzer, Cathode Ray Oscilloscope, Multimeter, etc.
- Strong Knowledge in using the PIM master Analyzer and Spectrum Analyzer
- Having Research & development experience in designing new antennas
- Prepared Bill of Materials (BOM) & complete technical reports

SKILLS:

RF Antenna Design and test, PCB Design, CST Microwave Studio, HFSS, PSpice, 2G, 3G, 4G, 5G, GSM, GPS, CDMA, WCDMA, LTE, AutoCAD, Bluetooth, ADS, MATLAB

Education:

University of South Alabama, Mobile, AL, USA	GPA 4.0/4.0	December 2017
M.Sc. in Electrical Engineering		

Jawaharlal Nehru Technological University, Hyderabad, TS, India	GPA 3.5/4.0	July 2013
B.Tech. in Electronics & Communication Engineering		

Professional Experience:

Employer: TTS/Amdocs Company, Frisco, Texas, USA	May 2018 – Present
Job Title: RF Engineer	

- Develop test strategies and procedures for Passive Intermodulation and provide solution for fault analysis
- Examine radiation patterns and response of the antenna system
- Conduct measurements and analyze performance of RF Antennas
- Measured scattering parameters S11 and S21, RF power, Noise floor, Intermodulation
- Reduce Noise floor interference and resolve the uplink RSSI levels
- Multi Band Port Interference Optimization data analysis, recommendation and resolution
- Mitigate the Passive Intermodulation sources using different RF barriers
- Perform in depth troubleshooting which has a high diversity imbalance and resolve issues causing PIM
- Worked with RF design team and RF super-components team while designing and testing antennas, filters and switches
- Hands on experience on Spectrum Analyzer, Network Analyzer, Passive Intermodulation Analyzer and Cathode Ray Oscilloscope
- Prepared a technical report with complete analysis and resolution

University of South Alabama, Mobile, AL, USA	January 2016 – December 2017
--	-------------------------------------

Advisor: Dr. Saeed I. Latif

Master's Thesis: Humanoid breast phantom sensing with Infrared sensors exposed to an ultra-wideband planar antenna

Job Title: Graduate Research Assistant

- Designed a planar ultra-wideband elliptical ring/slot monopole antenna with humanoid cubical and cylindrical breast phantoms for breast imaging by using CST microwave studio
- Examined the reflection coefficient and transmission coefficient of ultra-wideband planar antennas with breast phantom operating at 1.5GHz to 7.0GHz
- Tested an ultra-wideband planar antenna with humanoid cubical and cylindrical breast phantoms using Anritsu/Wiltron Vector Network Analyzer (VNA) (40MHz-40GHz)
- Studied the effects on the reflection coefficient and transmission coefficient of the antenna due to the presence of the humanoid breast phantom

- Prepared benign, malignant and fibrograndular phantoms in Applied Electromagnetics Research Lab (AERL) at the University of South Alabama
- Prepared research papers for journal & magazines and international conferences

Bharat Electronics Limited, Bengaluru, KA, India

January 2014 – January 2015

Job Title: Graduate Apprenticeship

- Designed L slot based wideband 4G LTE antennas for wireless system by using HFSS
- Tested RF modules on VNA (10MHz-26.5GHz)
- Examined gain, insertion loss, isolation loss, VSWR, etc. for RF modules
- Made PCB panel in Advanced Design System (ADS)
- Performed environment stress screening for sub-modules
- Good knowledge on CRO, DSO, Multimeter, Spectrum Analyzer
- Maintained new part number and uploading part list in SAP
- Documented project on SAP, MS-OFFICE, MS-EXCEL

Air voice Technologies, Hyderabad, TS, India

July 2013 - December 2013

Job Title: Industrial Automation Training

- Worked on Allen Bradley PLC and Mitsubishi PLC
- Configured hardware and software using PLC
- Developed software using automation system
- Coordinated with design engineer regarding refineries for power system
- Created and applied specifications by embedding software design programs.

ACADEMIC HONORS AND AWARDS:

- University of South Alabama Tuition Fellowship - Fall 2016, Spring and Fall 2017
- Secured highest GPA 4.0/4.0 in Electrical engineering department at University of South Alabama
- Graduated with highest distinction in Electronics & Communication engineering at JNTU Hyderabad
- Member of Tau Beta Pi Engineering Honor society
- Student member of Institute of Electrical and Electronics Engineers (IEEE)
- First prize in Death Race (Robot) at Indian Institute of Technology (BHU) Varanasi
- Second prize in Technical quiz competition at JNTU Hyderabad
- Excellence certificate from Robotics Academy of USA

ENGINEERING PROJECTS:

University of South Alabama, USA

Phantom sensing with infrared sensors to measure temperature rise

The goal of this project was to experimentally determine the temperature rise in benign and malignant phantoms.

Jawaharlal Nehru Technological University, India

Bus monitoring system based on Zig bee and GSM

The project mainly consists of Zig bee & GSM modem. The aim of the project was to solve the problem of monitoring the time of bus's arrival on and leaving the station and the problem of automatically reporting bus station accurately.

CONFERENCE PAPER:

F. Mansoor, T. Tan, Saeed I. Latif, "The performance of an Ultra-wideband elliptical ring monopole antenna with a humanoid breast phantom," Proc. 2017 IEEE AP-S International Symposium, and USNC/URSI National Radio Science Meeting (APS/URSI 2017), San Diego, California, USA, July 9-14, pp. 105-106, 2017

JOURNAL PAPER:

F. Mansoor and Saeed I. Latif, "A Planar Elliptical Slotted Patch Antenna without any Matching Liquid for Biomedical Application," International Journal of Bioautomation, vol. 24, no. 2, pp. 199-208, 2020

TECHNICAL POSTER AND PRESENTATION:

F. Mansoor, Saeed I. Latif, "Benign and Malignant Breast Phantom sensing with Infrared sensors exposed to an Ultra-wideband antenna", 5th Annual Nano Bio Summit, Atmore, AL, USA, November 9, 2017